

5(a)	current (through a conductor is directly) proportional to potential difference (across the conductor)	M1
	(provided that) temperature (of conductor remains) constant	A1
5(b)(i)	(ratio of) V/I increases (as p.d. increases)	B1
5(b)(ii)	(as p.d. increases, current increases so) temperature increases	B1
5(c)(i)	$I = 1.55 \text{ A}$	A1
5(c)(ii)	$P = VI$ or $P = I^2 R$ or $P = V^2 / R$	C1
	$= 6.0 \times 1.55 \times 2$ or $1.55^2 \times 3.87 \times 2$ or $(6.0^2 / 3.87) \times 2$ $= 19 \text{ W}$	A1
5(c)(iii)	$I = 1.78 - 1.55$ (= 0.23 A)	C1
	$R = 12.0 / 0.23$ $= 52 \Omega$	A1
5(d)	lamp P: p.d. across lamp decreases to zero so goes 'out'	B1
	lamp Q: p.d. across lamp increases to 12 V so gets brighter	B1